Solution:

Calculation for median follows by the following table

X	f	cf
0-10	20	20
10-20	25	45
20-30	32	77
30-40	40	117
40-50	42	159
50-60	35	194
60-70	10	204
70-80	8	N=212

Table : 8.18

$\frac{N}{2} = \frac{212}{2} = 106$. Class interval corresponding to cumulative frequency 106 is (30 – 40). So, the corresponding values from the median class are $L = 30$, $pcf = 77$, $f = 40$ and $c = 10$.

$$\text{Median} = L + \left(\frac{\left(\frac{N}{2}\right) - pcf}{f} \right) \times c$$

$$\text{Median} = 30 + \left(\frac{106 - 77}{40} \right) \times 10$$

$\therefore$ Median =37.25 (corrected to two places of decimals)

Calculations proceeded for mean deviation about the median.

X	f	M	$\lvert D\rvert = \lvert X - 37.25\rvert$	$f\lvert D\rvert$
0-10	20	5	32.25	645
10-20	25	15	22.25	556.25
20-30	32	25	12.25	392
30-40	40	35	2.25	90
40-50	42	45	7.75	325.5
50-60	35	55	17.75	621.25
60-70	10	65	27.75	277.5
70-80	8	75	37.75	302
	N=212			$\Sigma f \mid D \mid$=3209.5

Table : 8.19

Then the mean deviation about median is to be computed by the following

$$\text{MD about Median} = \frac{\Sigma f|D|}{N} = \frac{3209.5}{212} = 15.14$$

Coefficient of MD about Median

$$= \frac{\text{M.D. about median}}{\text{Median}} = \frac{15.14}{37.25} = 0.4064 = 0.41$$

(corrected to two decimal places).

NOTE

The above problem can also be solved for mean deviation about mean instead of median.

Exercise 8.1

1. Find the first quartile and third quartile for the given observations.
   2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22

2. Find Q_1 , Q_3, D_8 and P_{67} of the following data:

Size of Shares	4	4.5	5	5.5	6	6.5	7	7.5	8
Frequency	10	18	22	25	40	15	10	8	7

3. Find lower quartile, upper quartile, 7th decile, 5th decile and 60th percentile for the following frequency distribution.

Wages	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	1	3	11	21	43	32	9

4. Calculate GM for the following table gives the weight of 31 persons in sample survey.

Weight (lbs):	130	135	140	145	146	148	149	150	157
Frequency	3	4	6	6	3	5	2	1	1

08_11th_BM-STAT_Ch-8-EM.indd 194 21-04-2020 12:26:38 PM